John Francis Burkhart

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Education

Northern Arizona University *Flagstaff, AZ*
B.S. Mathematics, Summa Sum Laude. GPA: 4.0/4.0 *Aug. 2018 - May 2022*

Research Experience

North Carolina State University *Raleigh, NC*

Graduate Research Assistant *Aug. 2023 - April 2024*

- Co-authored and submitted paper (under review) on optimizing genome sampling for demographic and epidemiological inference with Markov Decision processes.
- Used Python, R, and bash scripts to prepare genetic data for analysis of pathogen phylogeny.
- Prepared medical image data for computer vision tasks (segmentation and classification).
- Completed two semesters of graduate coursework in genetics, statistics, and computer science.

Northern Arizona University *Flagstaff, AZ*

Research Technician *Aug. 2022 - Jul. 2023*

- Developed NIH-funded, Python-based open source epidemiological modeling software.
- Followed software engineering best practices by utilizing object-oriented design patterns, Git for version control, GitHub actions for CI/CD, and well-written documentation for project continuity.
- Mentored math and computer science undergraduate researchers.

Undergraduate Math Assistant *Aug. 2021 - May 2022*

- Studied the enumeration of signed permutations under the action of reversals and their applications to genome inversions.
- Solved several open problems, including finding a closed form solution for the number of signed permutations by reversal distance.
- Awarded $5,000 Hooper Undergraduate Research Award.
- Presented results at three different conferences.

Machine Learning Lab Research Assistant *Aug. 2021 - May 2022*

- Cleaned and implemented machine learning algorithms (Poisson regression, extreme gradient boosting, and random forests) on a large data set of multiplex polymerase chain reaction (mPCR) trials.
- Presented findings at undergraduate research symposium and published work to GitHub.

CUNY Baruch *Manhattan, NY (remote)*

Baruch Discrete Math REU Participant *Summer 2021*

- Adapted new methods for improving lower bound on Ramsey numbers to the multicolored bipartite setting.

Industry Experience

The CDC Foundation *Atlanta, GA (Remote)*

Data Analyst *Sep. 2024 - Present*

- Uses SQL, Python, and Tableau to extract, transform, load, and visualize non-profit fundraising data.
- Implements machine learning algorithms using Python (Pandas, Numpy, and Scikit-Learn) to aid in portfolio analysis and revenue forecasting.
- Automates time-consuming data quality and maintenance tasks using Python and Salesforce REST API.

ASM America *Phoenix, AZ (Remote)*

Intern Data Analyst *Summer 2022*

- Designed and built Power BI dashboards to visualize Global Product Team sales data

Teaching Experience

Northern Arizona University *Flagstaff, AZ*

Calculus I Peer Teaching Assistant *Aug. 2020 - Dec. 2020*
Linear Algebra Peer Teaching Assistant *Jan. 2020 - May 2020*
Peer Math Assistant (Calculus I and III tutor) *Jan. 2019 - May 2020*

Arizona Aspire Academy *Phoenix, AZ*

Math and Music Tutor *Aug. 2017 - May 2018*

Academic Writing

1. Optimizing genomic sampling for demographic and epidemiological inference with Markov decision processes. D. Rasmussen, M. Bursell, **F. Burkhart**. Currently under review. Preprint: *bioRxiv, 2025.06.30.662264*.
2. Enumerating signed permutations by reversal distance. F. Awik, **F. Burkhart**, H. Denoncourt, D.C. Ernst, T. Rosenberg, and A. Stewart (in preparation).

Presentations

2023

1. The Enumerative Combinatorics of Genome Inversions for Bioinformaticians. Invited talk given to the Phylodynamics Research Group at NC State.

2022

1. The Enumeration of signed permutations Under the Actions of Reversals. Virtual poster presentation at the Joint Mathematics Meeting.
2. The Enumeration of signed permutations Under the Actions of Reversals. Poster presentation at the Northern Arizona University Undergraduate Research Symposium.
3. The Enumeration of signed permutations Under the Actions of Reversals. Poster presentation at the Hooper Undergraduate Research Award Symposium.

2021

1. Improving the Bounds on Multicolored Bipartite Ramsey Number. Slide show presentation at the CUNY Discrete Math REU.
2. Using mlr3 for Machine Learning Research. Invited talk at the NAU Machine Learning Lab.

2020

1. Machine Learning Algorithms for Multiplex Polymerase Chain Reaction Primer Design. Virtual Poster presentation at the Northern Arizona University Undergraduate Research Symposium.

Awards and Honors

- NIEHS Environmental Health Bioinformatics (EHB) Fellowship *Aug. 2023*
- NAU Department of Mathematics and Statistics Outstanding Senior *May 2022*
- J. Harvey Butchart Mathematics Scholarship *Jan. 2021*
- Hooper Undergraduate Research Award (HURA) *Jan. 2021*

Technical Skills

Languages and Tools: Python, R, SQL, Tableau, Git, and LaTeX.